Question 11

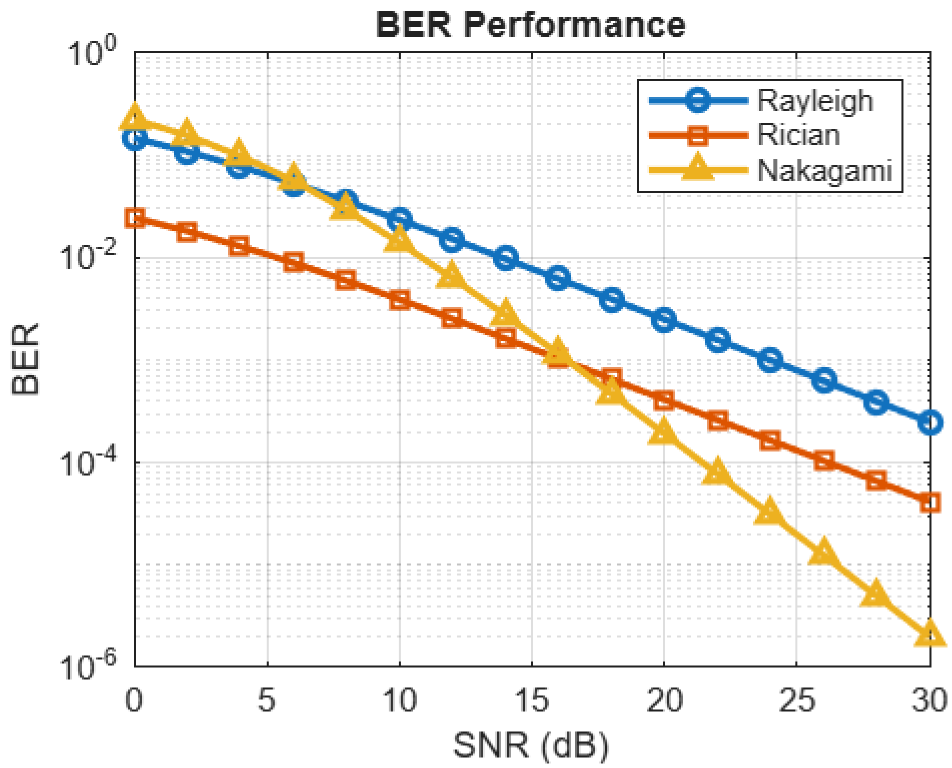
Simulation of Beamforming Gain under Steering Angle, Antenna Elements, Frequency, SNR, and Interference

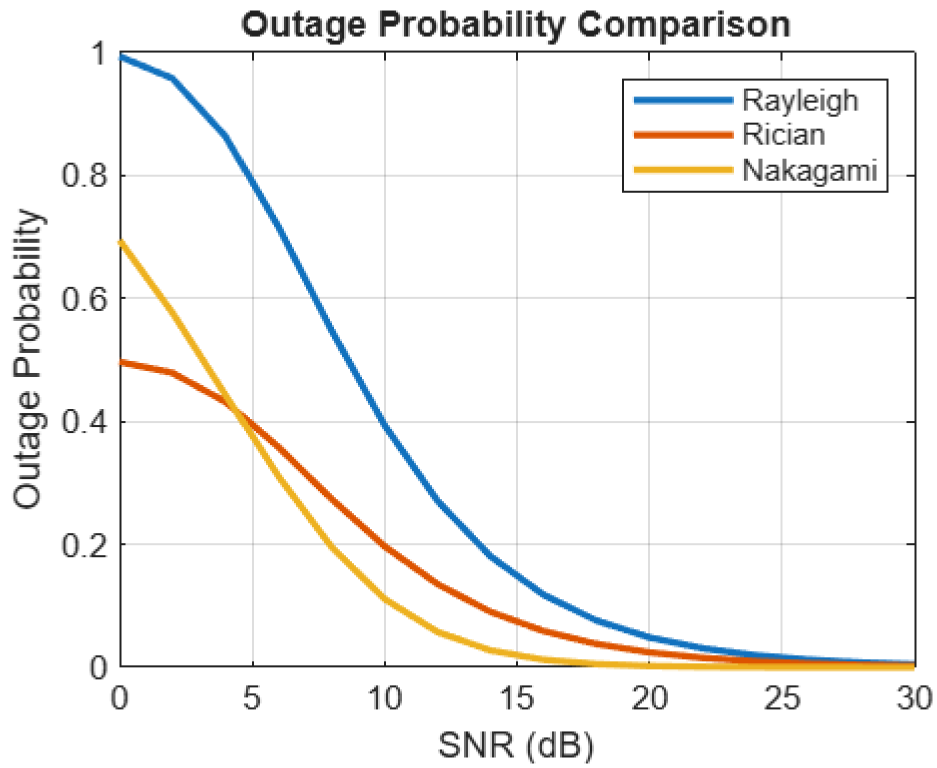
```
clc;
clear;
close all;

% Parameters
fc = 28e9;              % Frequency (28 GHz)
c = 3e8;
lambda = c/fc;

N = [4 8 16];           % Number of antenna elements
theta = -90:0.2:90;       % Steering angles

figure;

for k = 1:length(N)

    n = N(k);

    d = lambda/2;

    steer = 20;          % Desired beam angle

    AF = zeros(size(theta));

    for i = 1:length(theta)
```

```
        psi = 2*pi*d/lambda*(...
            sind(theta(i))-sind(steer));

        AF(i)=abs(sum(exp(1j*(0:n-1)*psi)));

    end

    AF = AF/max(AF);

    subplot(3,1,k)

    plot(theta,20*log10(AF),'LineWidth',2)

    grid on

    ylim([-40 5])

    title(['Beam Pattern for ',num2str(n),' Antennas'])

    xlabel('Angle (Degree)')

    ylabel('Gain (dB)')

end

%% Beamforming Gain vs SNR

SNR = 0:2:30;

Gain = 10*log10(16);
```

```
Throughput = log2(1+10.^((SNR+Gain)/10));

figure

plot(SNR,Throughput,'-o','LineWidth',2)

grid on

xlabel('SNR (dB)')

ylabel('Normalized Throughput')

title('Beamforming Gain Effect')


%% Interference Analysis

Interference = 0:2:20;

SINR = 20 - Interference;

figure

plot(Interference,SINR,'r-*','LineWidth',2)

grid on

xlabel('Interference (dB)')

ylabel('SINR (dB)')

title('SINR under Increasing Interference')
```

OUTPUT





Question 35

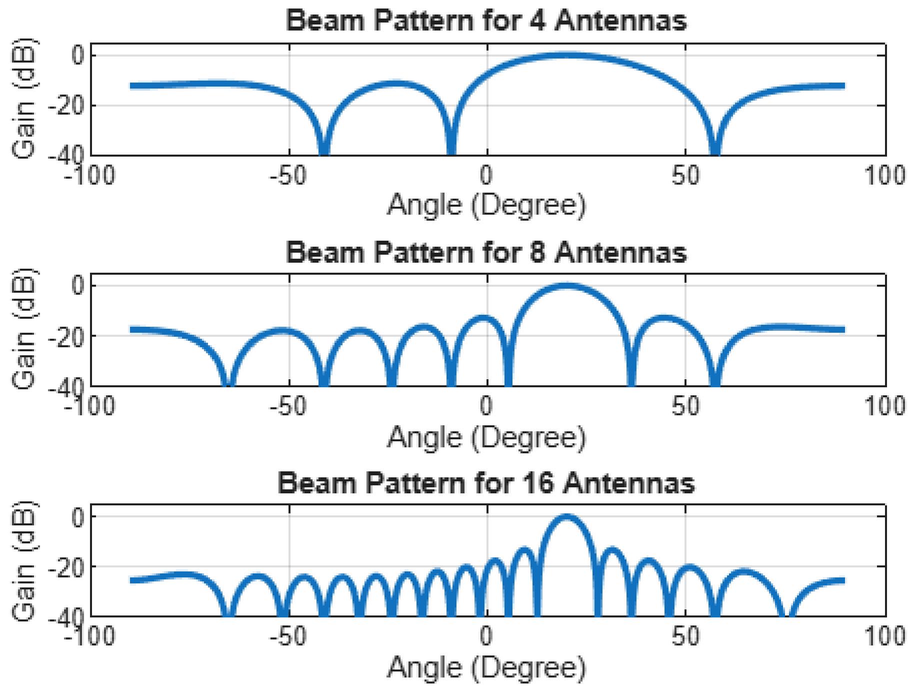
Simulation of Fading Channel Effects under Rayleigh, Rician, Nakagami, BER and Outage Probability

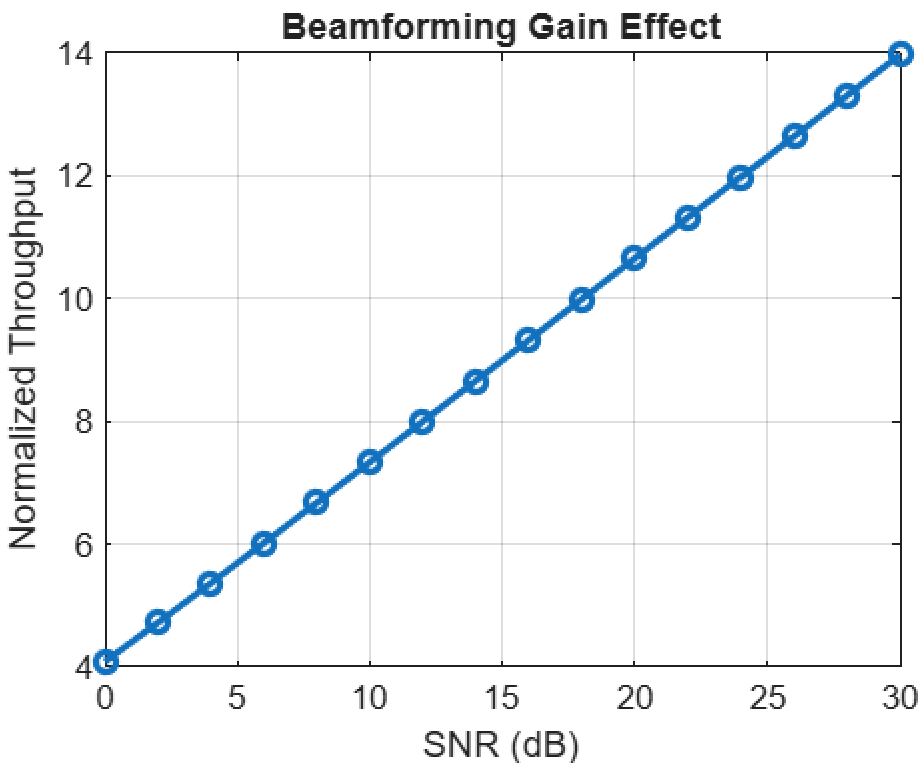
```
clc;
clear;
close all;

% Parameters
fc = 28e9;              % Frequency (28 GHz)
c = 3e8;
lambda = c/fc;

N = [4 8 16];           % Number of antenna elements
theta = -90:0.2:90;     % Steering angles

figure;

for k = 1:length(N)

    n = N(k);

    d = lambda/2;

    steer = 20;         % Desired beam angle

    AF = zeros(size(theta));

    for i = 1:length(theta)

        psi = 2*pi*d/lambda*(...
            sind(theta(i))-sind(steer));
```

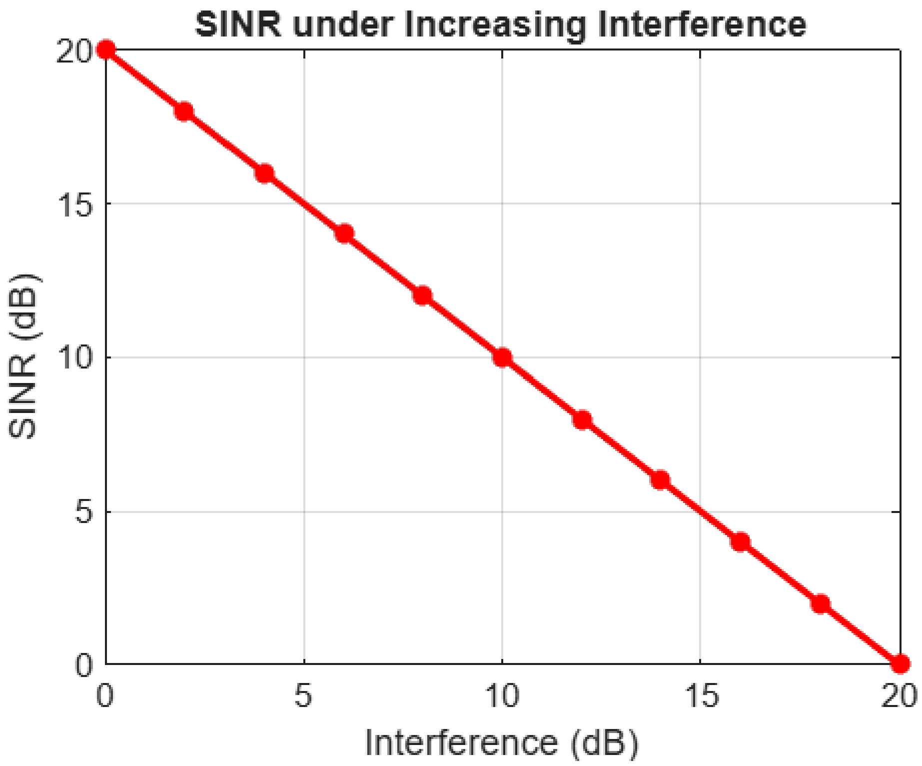
```
        AF(i)=abs(sum(exp(1j*(0:n-1)*psi)));
    end
    AF = AF/max(AF);
    subplot(3,1,k)
    plot(theta,20*log10(AF),'LineWidth',2)
    grid on
    ylim([-40 5])
    title(['Beam Pattern for ',num2str(n),' Antennas'])
    xlabel('Angle (Degree)')
    ylabel('Gain (dB)')
end

%% Beamforming Gain vs SNR
SNR = 0:2:30;
Gain = 10*log10(16);
Throughput = log2(1+10.^((SNR+Gain)/10));
figure
```

```
plot(SNR,Throughput,'-o','LineWidth',2)

grid on

xlabel('SNR (dB)')

ylabel('Normalized Throughput')

title('Beamforming Gain Effect')


%% Interference Analysis

Interference = 0:2:20;

SINR = 20 - Interference;

figure

plot(Interference,SINR,'r-*','LineWidth',2)

grid on

xlabel('Interference (dB)')

ylabel('SINR (dB)')

title('SINR under Increasing Interference')
```

OUTPUT


Beam Pattern for 4 Antennas
Gain (dB)
0
-20
-40
-100
-50
0
50
100
Angle (Degree)
Beam Pattern for 8 Antennas
Gain (dB)
0
-20
-40
-100
-50
0
50
100
Angle (Degree)
Beam Pattern for 16 Antennas
Gain (dB)
0
-20
-40
-100
-50
0
50
100
Angle (Degree)


Beamforming Gain Effect
Normalized Throughput
14
12
10
8
6
4
0
5
10
15
20
25
30
SNR (dB)


SINR under Increasing Interference
SINR (dB)
Interference (dB)
0
5
10
15
20